

	EMAX2 LV Circuit Breaker Routine Verifications	Doc.No.:
Factory Acceptance Check List		Doc.Ref:

Device Identification

Sales Order #		Rated Current	
Serial Number		Rated Voltage	
Apparatus Type		Control Voltage	

1. Visual Inspection

Item	Requirement	Result	Remarks
General data	Matched Drawing		
General condition	No damage		
Insulation	Clean / no tracking		
Breaker contacts	No visible wear or arcing marks		
Terminals	Secure & tight		

2. Mechanical Operations

Item	Requirement	Result	Remarks
Racking prevented while breaker is closed	Disconnect to Test		
Racking prevented while breaker is closed	Test to Disconnect		
Racking prevented while breaker is closed	Test to Connect		
Racking prevented while breaker is closed	Connect to Test		
Breaker removal prevented while closed.	Check		

3. Electrical Operations

Item	Requirement	Result	Remarks
Wiring Check (Point to point Check)	Matched Wiring Diagram		
Five Manual Operations completed successfully	Without Failure in between		
Five Opening operations	Shunt Trip Installed		
Two Operations completed successfully	Without Failure in between		
Five Charging Operations at minimum Voltage	Min. Vol = V DC		
Five Operations (Close/Open) at minimum Voltage	Min. Vol = V DC		
Five Charging Operations at maximum Voltage	Max. Vol = V DC		
Five Operations (Close/Open) at maximum Voltage	Max. Vol = V DC		
Two Anti-Pump Operations at normal voltage	Normal Vol.= V DC		

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4. Contact Resistance Test (Ductor Test)

Acceptance Criteria: $\leq 50 \mu\Omega@800A$, $\leq 40 \mu\Omega@1200A$, $\leq 30 \mu\Omega@1600A$, $\leq 20 \mu\Omega@3200A/4000A$ ($\pm 10\%$ phase-to-phase balance).

Phase	Measured ($\mu\Omega$)	Acceptance ($\mu\Omega$)	Result
Phase A (L1)			
Phase B (L2)			
Phase C (L3)			

5. Primary Current Injection – Trip Device Testing

Acceptance criteria applied per ETU (Electronic Trip Unit) settings.

Function	Set Point	Accept. Criteria	Phase A	Phase B	Phase C	Results
Long-time Pickup	1600 A	10%	1520	1520	1520	Pass
Long-time Delay*	3 Sec	10%	2.9	2.9	2.9	Pass
Short-time Pickup						
Short-time Delay**						
Inst. Pickup						
Inst. Delay**						
Ground Pickup						
Ground Delay**						

*The injected current should be 3 times the pickup value.

**The injected current should be 1.5 times the pickup value.

6. Dielectric Withstand Test (Hi-Pot)

Test Voltage: 2200 V/ 60 Hz for 60 seconds (each pole / each phase).

Acceptance Criteria: No flashover or dielectric failure.

Phase	Test Voltage	Duration	Result
Phase A – Ground	2200 V / 60 Hz	60 sec	
Phase B – Ground	2200 V / 60 Hz	60 sec	
Phase C – Ground	2200 V / 60 Hz	60 sec	

7. Control Wiring Hi-Pot Test

Acceptance Criteria: No flashover during specified duration.

Circuit	Test Voltage (AC)	Duration	Result	Remarks
Secondary Wiring – Ground	1500 VAC / 60 Hz	60 sec		Not installed
Charging Motor – Ground	1000 VAC / 60 Hz	60 sec		Not installed

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8. Additional Features & Options

Feature	Specification	Result	Remark
2nd Close Coil Operation			
2nd Close Coil Operation	At Minimum Voltage		Not installed
2nd Close Coil Operation	At Maximum Voltage		Not installed
Fan Operation			
Fan Operation	Activation		
Fan Operation	Runing for 60 Sec		
Undervoltage			
Max. Pickup Value	Voltage Setting=	V	
Max. Drop Out Value	Voltage Setting=	V	
Time Delay	Time setting=	Sec	

9. Documentation & Signatures

Wiring Dia-gram No.	Counter No.	Tested By	Signature	Date

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